

BEFA

Business Economics & Financial Analysis

Question Bank – Model Answers

Units 3, 4 & 5

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Q1. Discuss the Features of Capital Budgeting and its Proposals

Capital Budgeting is the process of planning and evaluating long-term investment decisions involving large capital expenditure. It helps a firm decide which projects or assets to invest in for maximum returns over an extended period.

Features of Capital Budgeting

- **Long-term decisions:** Involve long-term impact on the firm's profitability and financial position.
- **Large capital expenditure:** Significant funds are involved, making decisions crucial and difficult to reverse.
- **Risk and uncertainty:** Decisions about future operations are subject to uncertainty and business risk.
- **Irreversibility:** Most capital investment decisions cannot be easily reversed once implemented.
- **Time value of money:** Recognizes that money today is worth more than the same amount in the future.
- **Impact on future costs:** Capital investments significantly affect future operating costs and revenues.

Types of Capital Budgeting Proposals

- **Independent Proposals:** Acceptance or rejection of one project does not affect another project.
- **Mutually Exclusive Proposals:** Acceptance of one project means rejection of the other (e.g., choosing between two machines).
- **Contingent / Dependent Proposals:** Acceptance of one project depends on accepting another related project.
- **Replacement Proposals:** Decisions to replace old assets with new ones to improve efficiency.
- **Expansion Proposals:** Investment in new projects to expand existing business operations.

Q2. Explain Various Concepts of Accounting

Accounting concepts are the **fundamental assumptions and rules** that guide the preparation and presentation of financial statements. They ensure uniformity and reliability in accounting practice.

1. Business Entity Concept

The business is treated as a separate entity from its owner. All transactions are recorded from the business's perspective, not the proprietor's. The proprietor is treated as a creditor of the business.

2. Going Concern Concept

The assumption is that the business will continue to operate indefinitely unless dissolved for some reason. Assets are therefore recorded at cost, not liquidation value.

3. Money Measurement Concept

Only transactions that can be expressed in monetary terms are recorded. Non-monetary factors like employee morale or business reputation are not recorded.

4. Cost Concept

Assets are recorded at their original purchase cost. In the balance sheet, assets appear at cost less accumulated depreciation, not at current market value.

5. Accounting Period Concept

The life of a business is divided into specific time periods (usually one year) for the purpose of preparing financial statements and measuring performance.

6. Dual Aspect Concept

Every business transaction has two aspects — a debit and a credit. For every debit there is a corresponding credit. This forms the basis of double entry bookkeeping.

7. Matching Cost Concept

Expenses incurred in an accounting period must be matched against the revenue earned in that same period to determine the true profit or loss.

8. Realization Concept

Revenue is recognized only when a sale is actually made — when goods are transferred to the buyer and the buyer becomes legally liable to pay.

Q3. What are the Different Types of Accounts?

All business transactions fall into three categories — those relating to persons, property, and income/expenses. Accordingly, three types of accounts are maintained:

Type of Account	Rule	Examples
Personal Accounts (Transactions with persons)	Debit the Receiver Credit the Giver	Krishna's A/C, SBI A/C, Capital A/C, Drawings A/C
Real Accounts (Property / Assets)	Debit what comes in Credit what goes out	Cash A/C, Furniture A/C, Building A/C, Machinery A/C
Nominal Accounts (Expenses, Losses, Income, Extra)	Debit all expenses & losses Credit all incomes & gains	Salaries A/C, Rent A/C, Commission Received A/C

Before recording any transaction it is essential to identify which accounts are affected and apply the appropriate rule to determine what is debited and what is credited.

Q4. Explain Payback Period Method of Capital Budgeting with Formula and Examples

The **Payback Period** is the time required for the cash inflows from an investment to recover the original cost. It is one of the simplest and most widely used capital budgeting methods.

Formula (Uniform Cash Flows)

Payback Period = Initial Investment ÷ Annual Cash Inflow

When cash inflows are unequal, cumulative inflows are computed year by year until the investment is fully recovered.

Decision Rule

The project with the **shorter payback period** is preferred.

Example 1 – Uniform Cash Flows

Item	Amount
Initial Investment	Rs. 1,00,000
Annual Cash Inflow	Rs. 25,000
Payback Period	$1,00,000 \div 25,000 = 4$ Years

Example 2 – Unequal Cash Flows

Year	Cash Inflow (Rs.)	Cumulative Inflow (Rs.)
1	20,000	20,000
2	30,000	50,000
3	30,000	80,000
4	20,000	1,00,000 ← Investment Recovered

Payback Period = 4 Years

Merits & Demerits

- **Merits:** Simple to calculate, easy to understand, useful for liquidity analysis.
- **Demerits:** Ignores the time value of money; ignores cash flows occurring after the payback period.

Q5. Explain the Ratios Used in Financial Analysis with Formulas

Accounting ratios show inter-relationships among various accounting data. They are classified into four broad types:

A. Profitability Ratios

Ratio	Formula	Ideal
Gross Profit Ratio	$(\text{Gross Profit} / \text{Net Sales}) \times 100$	Higher is better
Net Profit Ratio	$(\text{Net Profit after Tax} / \text{Net Sales}) \times 100$	Higher is better
Operating Ratio	$((\text{COGS} + \text{Operating Exp.}) / \text{Net Sales}) \times 100$	Lower is better
Return on Capital Employed	$(\text{Operating Profit} / \text{Capital Employed}) \times 100$	Higher is better
Earnings Per Share	$(\text{Net Profit after Tax} - \text{Preference Dividend}) / \text{Number of Equity Shares}$	Higher is better

B. Turnover / Activity Ratios

Ratio	Formula	Ideal
Stock Turnover Ratio	$\text{COGS} / \text{Average Stock}$	Higher is better
Debtors Turnover Ratio	$\text{Credit Sales} / \text{Average Debtors}$	Higher is better
Creditors Turnover Ratio	$\text{Credit Purchases} / \text{Average Creditors}$	Lower is better
Working Capital Turnover	$\text{Sales} / \text{Working Capital}$	Higher is better
Fixed Assets Turnover	$\text{Sales} / \text{Fixed Assets}$	Higher is better

C. Liquidity / Financial Ratios

Ratio	Formula	Ideal Ratio
Current Ratio	$\text{Current Assets} / \text{Current Liabilities}$	2 : 1
Quick (Acid-Test) Ratio	$\text{Quick Assets} / \text{Current Liabilities}$	1 : 1
Absolute Liquid Ratio	$\text{Absolute Liquid Assets} / \text{Current Liabilities}$	0.5 : 1

D. Leverage / Solvency Ratios

Ratio	Formula	Ideal Ratio
Debt-Equity Ratio	$\text{Outsiders' Funds} / \text{Shareholders' Funds}$	2 : 1
Proprietary Ratio	$\text{Shareholders' Funds} / \text{Total Assets}$	1 : 3
Debt to Total Fund Ratio	$\text{Outsiders' Funds} / \text{Capital Employed}$	2 : 3

Q6. Explain Features, Advantages and Disadvantages of Cooperative Societies

A **Cooperative Society** is a voluntary association of persons, usually from economically weaker sections, who join together to achieve a common economic objective through collective working and mutual help.

Features

- **Voluntary Association:** Any person can join or leave freely at any time and withdraw his capital.
- **Spirit of Cooperation:** Works on the motto 'each for all and all for each' — members work for collective interest, not self-interest.
- **Democratic Management:** Follows 'one man, one vote' regardless of capital contributed, ensuring economic equality.
- **Capital:** Raised through share capital from members; cooperative banks may also provide loans.
- **Mutual Help:** Operates on the principle of mutual welfare; surplus is distributed as dividend per bye-laws.

Advantages

- Easy to form and register under the Cooperative Societies Act.
- Promotes economic equality and self-help among weaker sections.
- Democratic and transparent management ensures accountability.
- Limited liability protects members' personal assets.
- Government support and financial assistance available.

Disadvantages

- Limited capital due to lower income of members.
- Lack of professional management skills.
- Slow decision-making due to democratic voting structure.
- Possibility of internal conflicts and politics among members.
- Lack of secrecy as accounts are open to all members.

Q7. What are the Different Forms of Organization with Features?

1. Sole Trader

- Oldest and simplest one-man form of business.
- Owner bears all risk with his own capital, skill and intelligence.
- Easy to form and close; complete operational freedom.
- Unlimited liability; maximum secrecy; low taxation.
- Direct motivation — all profits belong to the sole trader.

2. Partnership

- Two or more persons share profits/losses in an agreed ratio.
- Governed by the Indian Partnership Act, 1932.
- Min. 2 partners; max. 10 (banking) / 20 (non-banking).
- Unlimited liability for all partners.
- Division of labour; flexibility; personal contact with customers.

3. Joint Stock Company

- Voluntary association of investors in share capital.
- Artificial legal person with separate legal existence.
- Limited liability — shareholders' personal assets are protected.
- Perpetual succession; transferability of shares; common seal.
- Professional management; democratic governance.

4. Co-operative Organisation

- Voluntary democratic association for mutual benefit.
- One man, one vote — regardless of capital contributed.
- Service motive dominates; surplus distributed as dividend.
- Capital raised from members and cooperative banks.

5. Government Company

- At least 51% paid-up capital held by Central or State Government.
- Registered under Companies Act, 1956; separate legal entity.
- Directors nominated by government; subject to ministerial control.
- Operates with administrative autonomy and financial freedom.

6. Private Company

- Restricts share transferability; prevents public subscription.
- Minimum 2, maximum 200 members.
- Must include 'Private Limited' or 'Pvt. Ltd.' in name.
- Enjoys several exemptions not available to public companies.

Q8. Explain the Internal Rate of Return (IRR) Method and its Importance

The **Internal Rate of Return (IRR)** is the discount rate at which the **Net Present Value (NPV)** of a project becomes zero, i.e., the present value of all future cash inflows exactly equals the initial investment.

Decision Rule

Condition	Decision
IRR > Cost of Capital (required rate)	ACCEPT the project
IRR < Cost of Capital	REJECT the project
Comparing multiple projects	Choose the one with the HIGHEST IRR

Interpolation Formula

$$\text{IRR} = \text{Lower Rate} + \left[\frac{\text{NPV at Lower Rate}}{\text{NPV at Lower Rate} - \text{NPV at Higher Rate}} \right] \times (\text{Higher Rate} - \text{Lower Rate})$$

Importance of IRR

- Considers the time value of money.
- Takes into account all cash flows over the entire life of the project.
- Provides a percentage return that is easy to compare with the cost of capital.
- Useful for ranking multiple investment proposals objectively.
- Helps management in accepting or rejecting projects based on true profitability.

Limitations

- Complex to calculate — requires trial and error or interpolation.
- May give multiple IRR values when cash flows change signs more than once.

Q9. Explain Features of Joint Stock Company with its Merits and Demerits

A **Joint Stock Company** is a voluntary association of persons who invest jointly in share capital. The word originates from Latin — 'com' (together) + 'pany' (bread), meaning people come together to earn their livelihood.

Features

- **Artificial Person:** Created by law; has no physical form but exists in the eyes of law — intangible and invisible.
- **Separate Legal Existence:** The company is distinct from its members; it can own assets, borrow funds, sue, and be sued independently.
- **Limited Liability:** Shareholders' liability is limited to the face value of shares held — personal assets are fully protected.
- **Capital Divided into Shares:** Total capital divided into small units enabling even small investors to participate.
- **Transferability of Shares:** Shares of a public company can be freely sold; private company shares cannot be transferred.
- **Common Seal:** Acts as the company's official signature on all legal documents.
- **Perpetual Succession:** The company continues to exist regardless of the death or insolvency of any member.
- **Winding Up:** Since law creates it, only law can dissolve it under special circumstances.
- **Name ends with 'Limited':** Indicates to outsiders that they are dealing with a limited liability entity.

Merits

- Mobilization of large resources from many investors.
- Limited liability encourages public investment and risk-taking.
- Perpetual succession ensures business continuity.
- Professional management due to availability of large funds.
- Enables growth and expansion through reserves and reinvestment.
- Encourages savings and investment habits among the public.

Demerits

- Complex and costly to form; requires formal registration.
- Greater government regulation and compliance obligations.
- Less secrecy — accounts must be published and disclosed.
- Risk of misuse of power by management (separation of ownership and control).
- Delay in decision-making due to democratic governance structure.

Q10. Compare Sole Trader and Partnership Business Organization

Aspect	Sole Trader	Partnership
Meaning	Single person owns & manages	Two or more persons jointly own & manage
Formation	Very easy; no legal formalities	Requires an agreement (oral or written)
Members	Only one	Min. 2; Max. 10 (banking) / 20 (non-banking)
Capital	Limited to owner's resources	More capital pooled from multiple partners
Liability	Unlimited — personal assets at risk	Unlimited — all partners personally liable
Management	Owner alone manages	Jointly managed; division of labour possible
Decision Making	Quick and prompt	May take time; requires consensus
Secrecy	Maximum — only one person	Somewhat less; secrets shared among partners
Continuity	Ends with death/insolvency of owner	May dissolve if a partner dies or retires
Risk	Borne entirely by one person	Shared proportionately among all partners
Taxation	Low income tax rates	Tax on each partner's share of profit
Flexibility	Very high	High; partners can jointly revise decisions

Conclusion: Sole tradership suits small businesses needing quick decisions and complete secrecy, while partnership is better for businesses that need more capital, expertise, and division of labour among likeminded persons.

Q11. Define Double Entry Book Keeping. Explain its Advantages and Disadvantages

Definition

The **American Institute of Certified Public Accountants (AICPA)** defines accounting as 'the art of recording, classifying and summarizing in a significant manner and in terms of money, transactions and events which are in part at least of a financial character, and interpreting the results thereof.'

Double Entry Book Keeping is based on the Dual Aspect Concept — every transaction has two aspects, a debit and a credit. For every debit entry there is a corresponding and equal credit entry, giving a complete record of all business transactions.

Advantages

- **Complete Record:** Both aspects of every transaction are recorded, giving a full picture of business affairs.
- **Detection of Errors:** The trial balance helps verify arithmetical accuracy; discrepancies reveal errors.
- **Prevention of Fraud:** Work divided among many persons with checks and counter-checks makes fraud difficult to conceal.
- **Preparation of Financial Statements:** Trading account, profit & loss account, and balance sheet can be easily prepared.
- **Comparison:** Current year performance can be compared with past years to identify trends.
- **Control over Assets:** Provides accurate values of cash, stock, debtors, and fixed assets.
- **Tax Matters:** Properly maintained records help in settlement of all tax matters with authorities.
- **Documentary Evidence:** Accounting records can be used as legal evidence in court, supported by authentic vouchers.

Disadvantages

- **Does not record all events:** Only financial transactions are recorded; non-monetary factors like human resources or location advantage are omitted.
- **Does not reflect current values:** Data is historical in nature — buildings and machinery may be worth far more than balance sheet values.
- **Personal judgment:** Estimates for depreciation, bad debts, and stock valuation depend on personal judgment and may not always be accurate.
- **Inadequate cost information:** Only overall profitability is shown; cost and profitability of individual products or departments are not provided.

Q12. Explain the Net Present Value (NPV) Method

Net Present Value (NPV) is a capital budgeting technique that calculates the difference between the present value of all future cash inflows and the initial investment (cash outflow). It is grounded in the concept of the **time value of money** — a rupee today is worth more than a rupee in the future.

Formula

NPV = Present Value of Cash Inflows – Initial Investment

Future cash flows are discounted at the firm's required rate of return (cost of capital).

Decision Rule

Condition	Decision
NPV > 0	ACCEPT — project earns more than required rate
NPV < 0	REJECT — project earns less than required rate
NPV = 0	Break-even; depends on other strategic factors
Comparing projects	Choose the one with the HIGHEST NPV

Illustrative Example

Initial Investment = Rs. 50,000 | Discount Rate = 10%

Year	Cash Inflow (Rs.)	PV Factor @ 10%	Present Value (Rs.)
1	20,000	0.909	18,180
2	25,000	0.826	20,650
3	20,000	0.751	15,020
Total PV			53,850
Less: Investment			50,000
NPV			3,850 (Positive → ACCEPT)

Merits of NPV

- Considers the time value of money comprehensively.
- Takes into account all cash flows over the project's entire life.
- Provides an absolute measure of profitability in rupee terms.
- Consistent with the goal of maximizing shareholder wealth.

Demerits of NPV

- Difficult to understand compared to simpler methods like Payback Period.
- Requires accurate estimation of future cash flows and the correct discount rate.
- Does not give a rate of return — only an absolute value, making cross-project comparisons harder.